

National Coverage Determination (NCD)

Magnetic Resonance Imaging

220.2

Links in PDF documents are not guaranteed to work. To follow a web link, please use the MCD Website.

Tracking Information

Publication Number

100-3

Manual Section Number

220.2

Manual Section Title

Magnetic Resonance Imaging

Version Number**Effective Date of this Version**

03/22/1994

Ending Effective Date of this Version

09/28/2009

Description Information

Benefit Category

Diagnostic Services in Outpatient Hospital

Diagnostic Tests (other)

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

A - General

1. Method of Operation

MRI, formerly called nuclear magnetic resonance (NMR), is a noninvasive method of graphically representing the distribution of water and other hydrogen-rich molecules in the human body. In contrast to conventional radiographs or computed tomography (CT)

scans, in which the image is produced by X-ray beam attenuation by an object, MRI is capable of producing images by several techniques. In fact, various combinations of MRI image production methods may be employed to emphasize particular characteristics of the tissue or body part being examined. The basic elements by which MRI produces an image are the density of hydrogen nuclei in the object being examined, their motion, and the relaxation times, and the period of time required for the nuclei to return to their original states in the main, static magnetic field after being subjected to a brief additional magnetic field. These relaxation times reflect the physical-chemical properties of tissue and the molecular environment of its hydrogen nuclei. Only hydrogen atoms are present in human tissues in sufficient concentration for current use in clinical MRI.

2. General Clinical Utility

Overall, MRI is a useful diagnostic imaging modality that is capable of demonstrating a wide variety of soft-tissue lesions with contrast resolution equal or superior to CT scanning in various parts of the body.

Among the advantages of MRI are the absence of ionizing radiation and the ability to achieve high levels of tissue contrast resolution without injected iodinated radiological contrast agents. Recent advances in technology have resulted in development and Food and Drug Administration (FDA) approval of new paramagnetic contrast agents for MRI which allow even better visualization in some instances. Multislice imaging and the ability to image in multiple planes, especially sagittal and coronal, have provided a flexibility not easily available with other modalities. Because cortical (outer layer) bone and metallic prostheses do not cause distortion of MR images, it has been possible to visualize certain lesions and body regions with greater certainty than has been possible with CT. The use of MRI on certain soft tissue structures for the purpose of detecting disruptive, neoplastic, degenerative, or inflammatory lesions has now become established in medical practice.

Indications and Limitations of Coverage

B - Nationally Covered Indications (Effective November 22, 1985)

Although several uses of MRI are still considered investigational and some uses are clearly contraindicated (see subsection D), MRI is considered medically efficacious for a number of uses. Use the following descriptions as general guidelines or examples of what may be considered covered rather than as a restrictive list of specific covered indications. Coverage is limited to MRI units that have received FDA premarket approval, and such units must be operated within the parameters specified by the approval. In addition, the services must be reasonable and necessary for the diagnosis or treatment of the specific patient involved.

The MRI is useful in examining the head, central nervous system, and spine. Multiple sclerosis can be diagnosed with MRI and the contents of the posterior fossa are visible. The inherent tissue contrast resolution of MRI makes it an appropriate standard diagnostic modality for general neuroradiology.

The MRI can assist in the differential diagnosis of mediastinal and retroperitoneal masses, including abnormalities of the large vessels such as aneurysms and dissection. When a clinical need exists to visualize the parenchyma of solid organs to detect anatomic disruption or neoplasia, this can be accomplished in the liver, urogenital system, adrenals, and pelvic organs without the use of radiological contrast materials. When MRI is considered reasonable and necessary, the use of paramagnetic contrast materials may be covered as part of the study. MRI may also be used to detect and stage pelvic and retroperitoneal neoplasms and to evaluate disorders of cancellous bone and soft tissues. It may also be used in the detection of pericardial thickening. Primary and secondary bone neoplasm and aseptic necrosis can be detected at an early stage and monitored with MRI. Patients with metallic prostheses, especially of the hip, can be imaged in order to detect the early stages of infection of the bone to which the prosthesis is attached.

Disc Disease Diagnosis (Effective March 22, 1994)

The MRI may also be covered to diagnose disc disease without regard to whether radiological imaging has been tried first to diagnose the problem.

Gating Devices and Surface Coils (Effective March 4, 1991)

Gating devices that eliminate distorted images caused by cardiac and respiratory movement cycles are now considered state of the art techniques and may be covered. Surface and other specialty coils may also be covered, as they are used routinely for high resolution imaging where small limited regions of the body are studied. They produce high signal-to-noise ratios resulting in images of enhanced anatomic detail.

C - Contraindications and Nationally Noncovered Indications

1. Contraindications

The MRI is not covered when the following patient-specific contraindications are present. It is not covered for patients with cardiac pacemakers or with metallic clips on vascular aneurysms. MRI during a viable pregnancy is also contraindicated at this time. The danger inherent in bringing ferromagnetic materials within range of MRI units generally constrains the use of MRI on acutely ill patients requiring life support systems and monitoring devices that employ ferromagnetic materials. In addition, the long imaging time and the enclosed position of the patient may result in claustrophobia, making patients who have a history of claustrophobia unsuitable candidates for MRI procedures.

2. Nationally Noncovered Indications

The CMS has determined that blood flow measurement, imaging of cortical bone and calcifications, and procedures involving spatial resolution of bone and calcifications, are not considered reasonable and necessary indications within the meaning of section 1862(a)(1)(A) of the Social Security Act, and are therefore noncovered.

D. Other

All other uses of MRI for which CMS has not specifically indicated coverage or noncoverage continue to be eligible for coverage through individual local contractor discretion.

(This NCD last reviewed September 2004.)

Cross Reference

Also see NCDs on Magnetic Resonance Spectroscopy (§220.2.1) and [Magnetic Resonance Angiography](#) (§220.3).

Claims Processing Instructions

[TN 1831 \(Medicare Claims Processing\)](#) 

[TN 1998 \(Medicare Claims Processing\)](#) 

[TN 2171 \(Medicare Claims Processing\)](#) 

[TN 2293 \(Medicare Claims Processing\)](#) 

[TN 2307 \(Medicare Claims Processing\)](#) 

[TN 4147 \(Medicare Claims Processing\)](#) 

Transmittal Information

Transmittal Number

21

Coverage Transmittal Link

<https://www.cms.gov/Regulations-and-Guidance/Guidance/Transmittals/Downloads/R21NCD.pdf> 

Revision History

09/2004 - Made clerical/technical edits/clarifications with no substantive revisions and no changes to existing NCD policy. Effective date NA. Implementation Date 09/10/2004. ([TN 21](#) ) (CR 3425)

03/1994 - Provided coverage for diagnosis of disc disease without regard to whether radiological imaging has been tried first to diagnose problem. Effective date 03/22/1994. (TN 69)

02/1991 - Provided coverage of surface coils used to enhance image, and gating devices used to compensate for motion of body parts when used in conjunction with MRI. Effective date 03/04/1991. (TN 45)

07/1988 - Deleted noncoverage of paramagnetic contrast agents. Effective date 07/22/1988. (TN 28)

11/1985 - Set forth conditions under which MRI may be covered. Effective date 11/22/1985. (TN 1)

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- Original Consideration for Magnetic Resonance Spectroscopy for Brain Tumors (CAG-00141N) [↗](#)
- First reconsideration for Magnetic Resonance Imaging (MRI) (CAG-00399R) [↗](#)

Additional Information

Other Versions

Title	Version	Effective Between	
Magnetic Resonance Imaging	6	04/10/2018 - N/A	View
Magnetic Resonance Imaging	5	07/07/2011 - 04/10/2018	View
Magnetic Resonance Imaging	4	02/24/2011 - 07/07/2011	View
Magnetic Resonance Imaging	3	06/03/2010 - 02/24/2011	View
Magnetic Resonance Imaging	2	09/28/2009 - 06/03/2010	View
Magnetic Resonance Imaging	1	03/22/1994 - 09/28/2009	You are here

